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Toward Sustainability in the Built Environment: An Integrative Approach

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Introduction

The built environment refers to the human-made surroundings within which people reside and work, including buildings, green spaces, streets, urban centers, and critical infrastructure like water and energy networks. This sector holds a pivotal place in the global economy, contributing to approximately 10 percent of the global gross domestic product (1). However, its importance extends far beyond monetary metrics, exerting a profoundinfluence on emissions, resource utilization, and land use transformations across various industries. Notably, this sector stands responsible for more than one-third of global final energy consumption, underscoring its substantial contribution of nearly 40% to global energy-related greenhouse gas emissions while consuming nearly 60 percent of the world's total raw materials (1, 2). The recent surge in population and urbanization has exacerbated diverse challenges associated with the built environment, such as climate change, pollution, resource depletion, energy inefficiency, and economic instability (3). Addressing these multifaceted issues is crucial for fostering sustainability within the built environment.

Accelerating the transition toward creating built environments that are circular, well-balanced, inclusive, and resilient, while achieving net-zero emissions, stands as a paramount necessity. This shift is vital to accommodate the future spatial requirements and combat global warming, particularly in light of the rapid urbanization observed in developing nations. One effective strategy for advancing sustainability within the built environment involves material conservation through engineering approaches. A key facet of this approach involves the judicious utilization of engineered waste materials, such as waste plastics, recycled concrete aggregates, recycled asphalt, among others. These materials can be efficiently employed, accompanied by streamlined resource management practices, to mitigate resource depletion and curtail the release of toxic pollutants typically associated with conventional material production. Furthermore, the customization of design and engineering processes for adsorptive waste materials, exemplified by zeolites, can play a pivotal role in removing gaseous pollutants present in the vicinity. This comprehensive approach is essential for addressing the multifaceted challenges posed by resource scarcity and accumulating waste materials, while simultaneously nurturing sustainable development (4).

Therefore, that the development of practical methods and tools to facilitate the incorporation of resource conservation and recycling into the establishment of sustainable built environments becomes an urgent imperative. The objective of this special issue is to demonstrate the myriad opportunities for enhancing the sustainability and health of the built environment. This is accomplished through the presentation of novel perspectives, data, models, frameworks, and innovative approaches designed to prevent or mitigate long-term unintended consequences. The overarching goal is to integrate sustainable practices into the fabric of the built environment and actively contribute to the global pursuit of carbon neutrality. By advancing strategies and approaches conducive to

sustainable built environments, we strive to deepen our understanding of the most favorable pathways to toward achieving low-carbon development. Moreover, these endeavors bolster resilience in the face of the profound challenges posed by the rapid onset of climate change. This special issue comprises 21 articles, collectively illuminating the vast potential to enhance the sustainability and health of the built environment. These articles introduce fresh insights, new datasets, nuanced models, and robust frameworks, all dedicated to fortifying the sustainability and health aspects of the built environment. Additionally, this special issue serves as a platform for highlighting innovative approaches geared toward averting or mitigating undesirable long-term consequences. Notably, these articles coalesce around four central themes: 1) Materials Aspects of Sustainability, 2) Management Aspects of Sustainability, 3) Technology Aspects of Sustainability, and 4) Health Aspects of Sustainability. Overall, this special issue reflects the most recent scientific advances in constructing a more sustainable, carbon-neutral future.

Materials Aspects of Sustainability

As shown in this special issue, the incorporation of sustainable materials is of paramount importance for improving the sustainability of the built environment. The selection of materials characterized by reduced environmental impact, such as those sourcede from recycled or renewable resources, coupled with the adoption of practices that minimize carbon emissions across their entire lifecycle, emerges as a potent strategy. This concerted approach plays a substantive role in abating the ecological footproint of construction activities. Such practices markedly propel the progress toward greener and more sustainable practices in the construction industry, aligning closely with the imperative for environmentally responsible and socially conscious development.

Cement, a fundamental constituent of concrete, assumes a pivotal role in sustainability. With global cement production reaching an annual output of four gigatons and being a substantial source of CO_2 emissions, addressing its environmental impact emerges as a crucial imperative for forstering sustainable construction practices. The quest to minimize cement's carbon footprint constitutes an indispensable facet in the drive toward a greener built environment. Thiedeitz et al. undertook a noteworthy endeavor by replacing 25% of concrete cement with fine rice husk ash (FRHA). This substitution results in increased microstructure density, thereby yielding improvements in mortar durability, capillary suction, and carbonation. Notably, FRHA exhibited superior pozzolanic attributes compared to fly ash, culminating in higher compressive strength (2). Zhou et al. delved into the prospect of substituting cement in concrete with calcined engineering sediment waste, a byproduct of construction excavations. The viability of such substitution hinges on the calcination of engineering sediment waste at 700°C for 2 hours, a process that activates its pozzolanic potential. Concrete incorporating 30% calcined engineering sediment waste demonstrated a flexural strength of 4.3 MPa and a compressive strength of 43.4 MPa (5). Meanhwhile, Li et al. introduced high-purity diatomite (DE), a biosilica, can replace 30% of cement as a sustainable and regenerative byproduct of diatom-based biofuel production. DE biosilica exhibited the remarkable capacity todensify the interfacial transition zone within concrete. This effect was achieved through the consumption of portlandite and pore-filling via pozzolanic reactions. Consequently, these pozzolanic reactions led to an 80% reduction in capillary pore size, albeit accompanied by an increase in gel pore volume (6). As an alternative to cement, Sun and Yi harnessed ground granulated blast-furnace slag (GGBS) as a binder along with fine incineration bottom ash (IBA) and refuse marine clay (MC) as concrete aggregates. This composite formulation proved sufficiently robust for use as a construction material. Impressively, GGBS as a binder within a composition containing IBA-MC surpassed the performance of Portland cement in terms of strength characteristics (7). Lastly. MolaAbasi et al. explored the replacement of cement with zeolite, a natural pozzolan. to stabilize clay soil. Their investigation revealed that a 15% substitution of cement with zeolite yielded optimal results in terms of achieving maximum compressive strength and unconfined compressive strength (8).

With asphalt pavements covering approximately 90 percent of the world's paved roads and the extensive use of asphalt as a sealant in the built environment, the imperative arises for eco-friendly methods capable of enhancing asphalt's performance 'while mitigating its enviromental impact (9). Using waste material to improve asphalt pavement performance represents a promising avenue. Notably, Wang et al. found that the inclusion of waste polyethylene plastic results in a reduction in penetration and an increase in softening and Fraass breaking point, albeit exerting limited influence on rheological properties (10). Interestingly, the utilization of both waste plastics and jet grouting waste exhibited a positive impact on asphalt mastic. The incorporation of jet grouting waste into neat bitumen, for instance, increased the phase angle of asphalt mastic to 80°, making it stiffer. Meanwhile, the introduction of waste plastics increased bitumen's G' (shear modulus) by 183%. Notably, the incorporation of waste plastics into asphalt mastic yielded a 13% reduction in production energy and fuel usage compared to limestone powder, thereby prolonging the service life of the pavement (11). Furthermore, the incorporation of crumb-rubber-modified asphalt binder, aside from improving the high-temperature performance of asphalt pavements, demonstrated the potenital to mobilize the rubber component to better bind to virgin aggregates when

subjected to high temperatures ranging from 160°C to 190°C (12). To facilitate rubber modification of asphalt using crumb , Lyu et al. functionalized crumb rubber with waste vegetable oil . This bio-modified crumb rubber could be rendered conducive for deoxidation and devulcanization, facilitating the release of rubber components into asphalt binder. Notably, this bio-modified crumb rubber exhibited the capacity to increase crack resistance by mitigating the phase separation between rubber particles and asphalt binder under low-temperature conditions (13). In a separate investigation, Zakertabrizi et al. ventured into the utilization of waste cooking oil as a partial replacement for asphalt binder, followed by the incorporation of waste sulfur to enhance the bio-modified asphalt. The resultant modified bio-oil/high-sulfur bitumen, following thermal curing at 180°C for 30 minutes, demonstrated an impressive threefold increase in flexibility compared to control bitumen. However, it's worth noting that uncured sulfur exhibited a diminishing effect on flexibility, persiting even after 60 days of ambient curing (14).

To mitigate the construction industry's carbon footprint, constructors can explore the potential of polymers and polymer nanocomposites, which offer customizable attributes for creating lightweight, multifunctional, high-performance alternatives to traditional building materials (15). An innovative approach suggested by Fabiani et al. involves repurposing the vast quantities of disposable surgical masks generated during pandemics for the production of wall panels endowed with acoustic and heat-insulating properties. Tests conducted on these wall panels, with a thickness of 3cm, fabricated from either shredded or intact masks, yielded promising results. The thermal conductivity values were measured at 0.0425 W/mK for shredded masks and 0.0494 W/mK for intact used masks, indicating efficient thermal insulation. Additionally, the acoustic absorption coefficient for these panels was northworthy, with a value of 0.70 recorded for used blue masks across the frequence range of 4050 Hz to 6400 Hz. Furthermore, shredded masks exhibited even higher accoustic absorption coefficients, ranging from 0.90 to 0.99 within the frequency range of 5282 Hz to 6400 Hz. Notably, the sound-absorption capabilities were found to increase proportionally with material thickness (16).

Management Aspects of Sustainability

As exemplified in this special issue, the integration of sustainability into the built environment necessitates a comprehensive approach to addressing three key components of management. Firstly, resource and waste management must strive for optimal output by minimizing input waste. This encompasses optimizing material selection and design to minimize waste generation during both construction and demolition phases (17, 18). Strategies aimed at reducing the consumption of raw materials include using recycled materials, promoting zero-waste construction, and adopting green alternative construction materials (19, 20).

Secondly, strategic planning and design must sustainability from a project's inception, such as site selection, land use, and transportation networks. For example, in the building industry, the development of a holistic life-cycle sustainability grading tool holds the potential to improve integrated design practice, optimize resource utilization, and enhance the transparency of sustainable decision-making (1).

Thirdly, continuous monitoring and evaluation processes are imperative to facilitate informed decision-making and necessary improvements. For instance, an in-depth analysis spanning technological, organizational, environmental, and financial facets of a project has underscored the importance of organizational variables in the formulation of sustainable infrastructure policies. Notably, organizational factors such as organizational culture, management support, and stakeholder demand have the greatest influence on promoting sustainable infrastructure. Consequently, the development of sustainable infrastructures should consider these drivers: technological drivers such as modular construction, lean supply chain management, innovations in information and communication technology, and the adoption of Internet of Things technology; environmental drivers such as market differentiation, environmental pressures, and legislative requirements; and financial drivers: budget adequacy, cost reduction initiatives, and financial incentives (21). By addressing these aspects of management, substantial progress can be achieved in successfully integrating sustainability into our built environment, thereby creating a future that is more resilient and environmentally responsible.

Technology Aspects of Sustainability

This special issue underscores the pivotal role of technology in driving sustainability within the built environment, facilitating innovative solutions, and enhancing environmental performance. Notably, advancements in energy-efficient technologies have been particularly influential in reducing both energy consumption and carbon emissions. One notable technology is automated construction, including 3D printing. This not only minimizes waste generation during construction but also has the potential to use diverse binders and recycled aggregates (19).

Moreover, technology-driven design and construction techniques such as Building Information Modeling (BIM) and virtual reality can enhance efficiency, minimize material waste, and improve project coordination. Methods

such as BIM-based life-cycle assessment models offer effective categorization and predictive capabilities concerning the quantity of building-demolition waste, thereby facilitating the selection of environmentally-friendly disposal options (17). Additionally, the application of 3D scanning for recycling waste wood and identifying suitable wood for reuse has yielded substantial reductions in the demand for new timber in construction (18).

The significance of technological progress extends to the development and implementation of sustainable materials and construction methods such as green-building surfaces and improved warm-mix asphalt pavements. Waste-derived superabsorbent polymers from the diaper industry have been used by Snoeck et al. to enhance the bio-receptivity of cementitious materials. These polymers, characterized by their high water-retaining properties, enable the creation of green concrete walls that support the colonization of microalgae, promoting green surfaces in urban areas (22). In a similar vein, Zou et al. explored the production of warm-mix asphalt using zeolite synthesized from sewage sludge ash. This innovative approach has exhibited improved characteristics compared to the use of commercial zeolite, resulting in reduced stiffness modulus and less environmental impact (23).

By leveraging technology in these various aspects, the built environment can attain enhanced sustainability, reduced environmental impact, and improved occupant well-being. The continued technological advancements and their integration into sustainable practices hold great promise in the quest for creating a more resilient and environmentally responsible built environment.

Health Aspects of Sustainability

This special issue underscores the pivotal role of health considerations in achieving sustainability within the built environment, emphasizing the creation of spaces that prioritize the well-being and health of occupants. To this end, health aspects must be considered throughout the life cycle of the materials, spanning production, use and recycling. One crucial aspect involves addressing pollution, which wields a significant impact on the health of city dwellers. By embracing low-emission building materials and reducing pollutants and contaminants, we have the capacity to cultivate healthier indoor environments.

For example, Li et al. conducted a study demonstrating the superior resistance to chloride and calcium leachate by partially replacing cement with high-purity diatomite, thus enhancing the durability of concrete structures (6). Similarly, using zeolite instead of cement for clay soil stabilization resulted in reduced energy consumption and reduced greenhouse gas emissions (8). Substituting jet grouting waste for limestone filler in the production of asphalt pavements has been shown to decrease nitrogen, sulfur, and phosphorus leakage into water by 85%, reducing human carcinogenic toxicity by 46% and freshwater eutrophication by 12% (11).

In addition to these measures, the incorporation of green spaces and microalgae on the surface of concrete buildings, in lieu of conventional gray facades, hold the potential to contribute to a more livable environment and bolster mental well-being (22). Similarly, nature-based and bio-based interventions furnish aesthetic value, improve air quality, and promote psychological restoration (24).

By affording paramount importance to health aspects in the design and development of sustainable built environments, we can enhance occupant health, well-being, and productivity, fostering thriving and sustainable communities. A holistic approach that simultaneously addresses both health and sustainability is indispensable for crafting a built environment that not only enhances our quality of life but also embodies a commitment to a sustainable future.

Concluding remarks

Comprising 21 articles, this special issue serves as a testament to the myriad opportunities for enhancing the sustainability and health of the built environment. These articles unveil innovative approaches that underscore the imperative of adopting a holistic perspective when assessing the enduring repercussions of contemporary recycling methodologies on both environmental sustainability and human health. This comprehensive compilation offers fresh insights, data, models, and frameworks, all converging toward the common goal of elevating the sustainability and health dimensions of the built environment.

Data Availability Statement

All data used in this study have been included in the manuscript.

Declaration of Competing Interest

The authors assert no competing financial interests or personal relationships that could have influenced the study's results.

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